

PROJECT GUIDE: AUTONOMOUS SOLAR CHARGER (MVP - PHASE 1)

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Area: Cyber-Physical Engineering / Autonomous Infrastructure

Project Focus: Intelligent Off-Grid Renewable Energy System

Status: Phase 1 (Bench Prototype) - Completed

1. PROJECT OVERVIEW: WHAT ARE WE BUILDING?

This project consists of developing a **modular off-grid solar energy infrastructure** (isolated from the traditional electrical grid). The goal of **Phase 1** is to build a robust and safe portable charger capable of harvesting solar energy, storing it efficiently in a lithium-ion cell, and providing a stable 5 V USB output to power small devices.

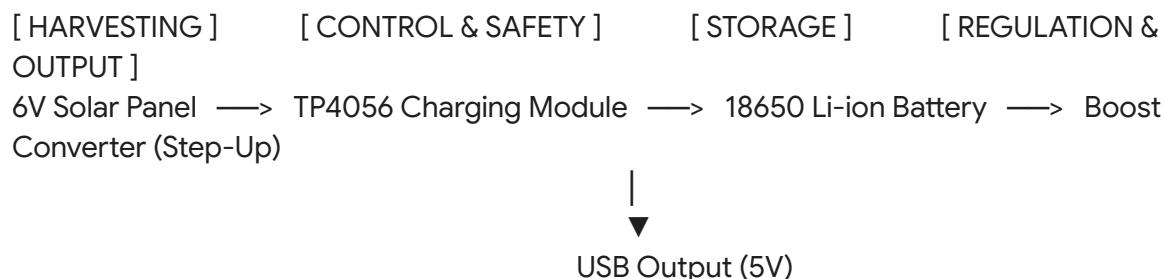
The key value of this project is not merely charging a phone, but rather **validating a functional block model** that can scale up to larger systems (such as sensor node telemetry, motorhome automation, or residential auxiliary backup power).

The Evolutionary Ecosystem (How this project scales):

- **Phase 1 (Current):** The Starting Module (Portable Charger and Mechanical/Electrical Protection).
- **Phase 2:** Automation & Optimization (Integrating an Arduino for active solar tracking).
- **Phase 3:** Scale & Power (A robust battery bank and inverter to supply residential AC power).

2. SYSTEM ARCHITECTURE: HOW DOES IT ALL CONNECT?

To make the technical presentation and energy flow easy to understand, the system is divided into four simple functional blocks:



The "Why" Behind Every Technical Decision:

A. The Solar Panel (6 V nominal instead of direct 5 V)

Solar panels are variable power sources that depend directly on light intensity. To charge a lithium battery up to its safe limit of 4.2 V , we need a slightly higher input voltage to compensate for the natural energy losses of the control board's internal components. A 6 V panel guarantees this voltage overhead even on partially cloudy days.

B. The Charge Controller (TP4056) - The Circuit's Security Guard

Lithium batteries are incredibly efficient, but highly sensitive to electrical abuse. If we attempt to feed power into them without control, they can overheat and suffer irreversible damage (with a risk of fire or explosion).

- **The Solution:** The TP4056 chip manages the charging process intelligently using the **CC/CV** (Constant Current / Constant Voltage) protocol. It feeds energy to the battery quickly and safely at first, and as the cell approaches its 4.2 V limit, the controller smoothly reduces the current until it cuts off the power entirely when the cycle is complete.

C. The Lithium Battery (18650)

We use the industrial standard 18650 cell (3.7 V nominal). It acts as our electrochemical fuel tank, storing the energy harvested during the day so it can be used at any time, even without sunlight.

D. The Boost Converter (Voltage Elevator)

A lithium battery's voltage fluctuates between 3.0 V (depleted) and 4.2 V (fully charged).

However, any modern USB device requires exactly a constant 5 V to charge without damaging its internal circuits.

- **The Magic:** The *Boost* converter takes the variable 3.7 V from the battery and, through a high-frequency switching circuit that stores energy in a magnetic inductor, elevates and stabilizes the output to exactly 5 V . This is far superior to older linear regulators, which wasted excess voltage as heat.

3. BILL OF MATERIALS (BOM) AND TOOLS

To bring this project to life with professional standards, we have organized our bench workshop setup, dividing it between durable long-term tools and the specific prototype components.

Prototype Components (The Circuit):

1. **Photovoltaic Solar Panel** ($5\text{ W} / 6\text{ V}$): Energy harvesting.
2. **TP4056 Charging Module with Integrated Protection**: Battery safety and charge manager.
3. **Heavy-Duty Battery Holder**: To house the cell securely (avoiding direct soldering onto the battery's metal caps).
4. **18650 Lithium-Ion Cell** (2500 mAh): Our energy storage accumulator.
5. **DC-DC Boost USB Module** (5 V): Output regulator to connect devices.

Bench Tools (Your Workshop):

- **Hikari HK-936B Soldering Station**: Allows precise temperature control at 320°C for perfect connections without damaging the electronic boards.
- **Minipa ET-1002 Digital Multimeter**: The engineer's "eyes," used to measure voltage, current, and test for short circuits.
- **Vonder Tool Case (110 pieces)**: Wire cutters, precision utility knife (with rotary safety lock), and screwdrivers for structural adjustments.
- **Antistatic Silicone Thermal Mat**: Protects the wooden workbench from burns and organizes small components.

4. ASSEMBLY AND COMMISSIONING STEP-BY-STEP

The assembly must follow an organized routine to ensure proper operation and mitigate any risk of short circuits.

Step 1: Preparation and Tinning

1. Turn on the soldering station to 320°C . Clean the tip of the iron and apply a small drop of solder to facilitate heat transfer.
2. Cut and strip about 3 mm of the conductive wires.
3. **Pro-Tip**: Apply a small amount of solder to the stripped wire tips and the board terminals before soldering them together (pre-tinning). This makes the final solder joint quick, clean, and mechanically strong.

Step 2: Soldering the Electrical Connections

1. **Battery Line**: Solder the wires from the 18650 battery holder to the **B+** (positive/red) and **B-** (negative/black) pads of the TP4056 module.

2. **Output Line:** Connect the **OUT+** pad of the TP4056 to the **IN+** input of the Boost converter. Repeat this process for the negative line (**OUT-** to **IN-**).
3. **Generation Line:** Solder the positive wire of the solar panel to the **IN+** pad of the TP4056, and the negative wire to the **IN-** pad.
4. **Insulation:** Slide heat-shrink tubing over all exposed splices to guarantee complete electrical insulation.

Step 3: Test Protocol (Before Connecting a Load)

With your multimeter in hand, execute these three safety verifications:

1. **Short-Circuit Check:** Set the multimeter to the continuity mode (audible beep). Touch the probes to the positive and negative lines of the circuit. **The device must not beep.** If it beeps, double-check and fix your solder joints before inserting the battery.
2. **Solar Generation Test:** With the solar panel exposed to direct sunlight, measure the input voltage at the TP4056. Verify that it reads around 6 V DC .
3. **Regulated Output Test:** Insert the 18650 battery into the holder. Measure the output terminals of the USB Boost converter. The display should show a stable value between 4.9 V and 5.1 V , proving that the system is ready to deliver power safely to your device.

5. FUTURE VISION: DESIGN SCALABILITY

The beauty of this MVP lies in its systemic expansion capacity. By mastering this core structure, we are fully prepared for the next strategic steps:

[MVP PHASE 1] —> [MVP PHASE 2] —> [MVP PHASE 3]
 Bench Solar Charger Automation & Tracking Off-Grid Power Scale
 (Harvesting + Charge + Boost) (Arduino + Motors + LDRs) (Inverter + Battery Bank)

1. **Mechanical Optimization (Solar Tracking):** By adding an Arduino and small servo motors coupled with light sensors, we will make the solar panel follow the sun autonomously. This can increase energy harvesting efficiency by up to 40% .
2. **Power Scalability:** By connecting battery cells in series and parallel configurations, and integrating a power inverter, we can elevate the output from Direct Current (12 V / 24 V DC) to Alternating Current (110 V / 220 V AC). This scales the prototype into a fully functional and independent residential off-grid power supply.